

Studying Loss Aversion and Sunk Cost effect in Gacha Games.

A Behavioural Economics Experiment in Game Design

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Gacha Games

Gacha games use a lottery-like mechanic where players spend in-game currency to obtain randomised rewards, such as rare characters or powerful equipment. This system, found in games like *Honkai Star Rail* and *Genshin Impact*, leverages unpredictability to engage and retain players, similar to slot machine gambling.

Wanna Give it a go?

But where does Behavioural Economics come in all this?



Literature Review.

Game Design Through the Lens of Behavioural Economics

Charles Butler links game mechanics to behavioral economics, highlighting how irrational behaviors like loss aversion drive player engagement. Using tactics like limited-time rewards and daily login incentives, his framework helps designers shape behaviors while raising ethical concerns about exploiting psychology in monetization, especially in gacha games.

When Prospect Theory Meets Optimal Pricing

A working paper by **Tan Gan** explores how gacha games use Prospect Theory, exploiting biases like probability weighting and loss aversion. Mechanisms like pity systems drive spending by balancing fairness and revenue, targeting naive and sophisticated players differently. The study raises ethical concerns about stochastic pricing and its regulatory implications.

Prospect Theory: An Analysis of Decision Under Risk

Kahneman and Tversky's Prospect Theory explains decision-making under risk, highlighting loss aversion and probability distortion. Players overweight rare events and evaluate outcomes relative to reference points. In gacha games, these biases drive addictive spending as players chase rare items or try to recover losses, defying rational decision-making.

The Psychology of Sunk Cost

Arkes and Blumer's study on the sunk cost effect reveals how past investments irrationally influence decisions. Controlled experiments showed individuals persist in failing ventures to justify prior costs, driven by loss aversion and commitment bias. In gacha games, this explains why players continue spending, unable to disregard prior unsuccessful attempts, amplifying irrational and addictive behavior.

Research Question.

"How do loss aversion and the sunk cost effect influence players' decision-making in Gacha games under conditions of risk and uncertainty?"

- How do sunk costs and loss aversion affect spending?
- Does the pity system (guaranteed reward after repeated attempts) influence player behavior?
- How does outcome framing (risk-averse vs. risk-loving) impact loss aversion?
- Does awareness of game mechanics (sophisticated vs. naive players) alter decisions?

Experiment Design

- **Task Setup:**
 - Participants receive 200 tokens and use them in a gacha-like system (10 tokens per pull).
 - Players can quit anytime.
 - Additional tokens available for "money" (not real money).
- Variations:
 - a. Disclosure of Pity System: Participants either informed or not about the pity system.
 - b. Outcome Framing: Risk-averse vs. risk-loving framing.

Data Collection

- Behavior Metrics:
 - Number of pulls, total tokens spent, and quitting point.
- Perceived Probabilities:
 - How players perceive chances of winning and the influence of the pity system.
- Emotional Responses:
 - Self-reported reasons for continuing or quitting.
 - Focus on regret, frustration, and satisfaction.

Gacha Game

Your current money: 80 points

Last spin result: Common

Common



Spin (Cost: 10)

Exit Game

Analysis Plan

- Independent Variables:
 - Disclosure of pity system (control vs. experimental).
 - Outcome framing (risk-averse vs. risk-loving).
- Dependent Variables:
 - Total spending, decision time, emotional responses.
- Quantitative Analysis: ANOVA or regression analysis.
- Qualitative Analysis: Survey responses for emotional insights

Hypotheses

- Hypothesis 1A: Participants aware of the pity system will spend more, as it mitigates loss aversion.
- Hypothesis 1B: These players will continue spending after low-value rewards, believing a high-value reward is due.
- Hypothesis 2: The pity system amplifies sunk cost effects, leading players to continue spending.
- Hypothesis 3A: Risk-averse framing leads to less spending.
- Hypothesis 3B: Risk-loving framing encourages more spending.
- Hypothesis 4: Pity system + risk-loving framing results in highest spending.
- Hypothesis 5A: Awareness of the pity system reduces frustration and increases satisfaction.
- Hypothesis 5B: Risk-averse framing increases regret and frustration.

Conclusion

Future Research Directions:

Investigate how changes in pull price affect behavior. Compare gacha game mechanics across different titles and their impact on player decisions. Study the influence of character inclusion (e.g., overly sexualized characters) on spending behavior.

Our experiment aims to provide empirical evidence of loss aversion and sunk cost effects in gacha games. By contrasting behavior in informed versus uninformed groups, we hope to demonstrate how psychological biases drive decision-making in uncertain gaming environments.

The findings may contribute to a better understanding of behavioral tendencies in digital entertainment and inform discussions on ethical game design and monetization strategies.

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